

Abstract

COGANT (Codebase-to-GNN Translation) converts software repositories into structured Active Inference artifacts expressed in the Active Inference Institute's **Generalized Notation Notation** (GNN)¹. It sits between classical program analysis, which already reasons over graphs such as call graphs and code property graphs [yamaguchi2014modeling], and data-driven methods that expect tensors, consistent node and edge vocabularies, and exportable training bundles [allamanis2018survey; wu2020comprehensive; scarselli2009graph].

The design centres on a **program graph IR** whose nodes and edges carry **confidence** and **provenance**. A **fixpoint translation engine** applies **19 declarative rules** (five structural, five semantic, two control, three behavioural, four resilience) that map nodes onto **7 Active Inference semantic roles** — **HIDDEN_STATE**, **OBSERVATION**, **ACTION**, **POLICY**, **PREFERENCE**, **CONSTRAINT**, and **CONTEXT** — with conflict resolution by priority and score. Each bundle includes a **Markov blanket partition** in $O(V + E)$ time. A **reverse synthesizer**